SVR ENGINEERING COLLEGE

(Autonomous) — Affiliated to JNTUA

SVREC ERP Portal

Enterprise Resource Planning System

Comprehensive Project Report

Project Name	SVREC ERP Portal
Institution	SVR ENGINEERING COLLEGE
Framework	Django 4.2 (Python)
Database	SQLite (Dev) / PostgreSQL (Production)
Report Date	04 April 2026
Version	v3.0 — Production Release

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1. Project Overview

1.1 Introduction

The **SVREC ERP Portal** is a full-featured, web-based Enterprise Resource Planning system purpose-built for SVR ENGINEERING COLLEGE (Autonomous). Developed using the Django framework (Python), this platform replaces manual and paper-based academic processes with a unified, role-based digital environment accessible from any device.

The system caters to three primary user roles — **HOD/Admin**, **Faculty/Staff**, and **Students** — each with a dedicated, feature-rich portal. The ERP covers the full academic lifecycle: student enrollment, attendance tracking, marks management, timetable scheduling, assignments, announcements, certificates, cloud storage, and advanced workflow automation.

1.2 Objectives

- Digitize all academic and administrative workflows of the institution.
- Provide role-based interfaces for Administrators, Faculty, and Students.
- Enable real-time attendance tracking, marks entry, and result computation.
- Support automated email notifications via a configurable workflow engine.
- Offer Excel import/export for student data and marks sheets.
- Provide a student cloud storage system for academic documents.
- Implement a certificate management system for student achievements.
- Generate academic calendars, timetables, and announcements.

1.3 Scope

This ERP is designed for a single college with multiple departments (branches), degree programs, academic regulations, and session years. The system supports multiple semesters (1-1 through 4-2) and sections (A, B, C). It is production-ready, deployed on PythonAnywhere and accessible at **syncx.pythonanywhere.com**.

2. System Architecture

2.1 Architectural Pattern

The application follows the **Model-View-Template (MVT)** architectural pattern — Django's adaptation of MVC. Business logic resides in Views (Python functions), data schemas are defined in Models (Django ORM), and presentation is handled by HTML Templates with Django template tags.

Layer	Technology	Responsibility
Presentation	HTML5, CSS3, Bootstrap 4, JS	User Interface & UX
Application	Django 4.2 (Python)	Business Logic, URL Routing, Views
ORM	Django ORM	Database Abstraction, Model Signals
Database	SQLite / PostgreSQL	Data Persistence
Authentication	Django Auth (AbstractUser)	Login, Role-Based Access
File Storage	Django FileField + Pillow	Profile Pics, Assignments, Cloud Files
Static Files	WhiteNoise	Serving CSS/JS in Production
Email	Django SMTP	Notifications & Workflow Emails
Excel	Pandas + OpenPyXL	Import/Export of Marks & Student Data

2.2 Application Structure

- college_management_system/ — Django project root (settings, URLs, WSGI/ASGI)
- main_app/ — Core Django application with all models, views, forms and templates
- main_app/views.py — Shared views: Login, Logout, Attendance API
- main_app/hod_views.py — Admin/HOD module (103 KB, ~2,500 lines)
- main_app/staff_views.py — Staff module (45 KB, ~1,200 lines)
- main_app/student_views.py — Student module (47 KB, ~1,300 lines)
- main_app/models.py — 25 data models defining the full relational schema
- main_app/forms.py — 30+ Django forms for all data-entry operations
- main_app/workflow_engine.py — Custom workflow automation engine
- static/ & staticfiles/ — CSS, JS, images (WhiteNoise served)
- media/ — User-uploaded files (profile pics, assignments, certificates)

3. Technology Stack

3.1 Backend

Component	Technology	Version / Notes
Web Framework	Django	≥ 4.2.0
Language	Python	3.x
WSGI Server	Gunicorn	≥ 20.1.0
Database ORM	Django ORM	Built-in
Database Driver	mysql-connector-python	≥ 8.0.0 (MySQL support)
URL Config	dj-database-url	≥ 0.5.0
Static Files	WhiteNoise	≥ 6.2.0
Image Processing	Pillow	Latest
HTTP Requests	requests	Latest
Data Processing	pandas	Latest
Excel I/O	openpyxl	Latest

3.2 Frontend

- HTML5 — Semantic markup for all 60+ templates
- CSS3 — Custom dark/glassmorphism design with transitions and animations
- Bootstrap 4.x — Responsive grid layout, modal dialogs, form controls
- JavaScript (ES6) — Dynamic interactions, AJAX calls, chart rendering
- Chart.js — Analytics dashboards with doughnut, bar, and line charts
- Drawflow.js — Visual drag-and-drop workflow builder canvas
- Select2 — Enhanced select dropdowns with search functionality
- Google Fonts (Inter) — Modern typography

3.3 Deployment

Hosting	PythonAnywhere (primary) | Local development: Django runserver
Live URL	https://syncx.pythonanywhere.com
Static Files	WhiteNoise middleware for production static serving
Database	SQLite for development, PostgreSQL-ready for production
Version Control	Git + GitHub (govardhan33699/svrecportalfianl)
Procfile	web: gunicorn college_management_system.wsgi

4. Data Models & Database Design

The ERP uses **25 relational models** managed through Django ORM. All models include timestamped audit fields (created_at, updated_at) and are linked via ForeignKey and OneToOneField relationships for referential integrity.

4.1 Core Models Summary

Model	Key Fields	Purpose
CustomUser	email, user_type, gender, profile_pic, address	Central user entity (HOD/Staff/Student)
Admin	admin (→ CustomUser)	HOD/Administrator profile extension
Staff	admin, course, designation, qualification, blood_group	Faculty profile with detailed attributes
Student	admin, course, roll_number, section, regulation, semester	Student profile with full academic details
Course	name, degree	Department / Branch (e.g., CSE, ECE)
Degree	name	Degree program (e.g., B.Tech, M.Tech)
AcademicLevel	name	Year level (1st Year, 2nd Year…)
AcademicSemester	name, academic_level	Semester within a year
Regulation	name, course, session	Academic regulation (e.g., R20, R23)
Session	start_year, end_year	Academic Year session
Subject	name, code, staff, course, semester, credits, max_marks	Subject / paper linked to staff & course
Attendance	subject, date, semester, session, period	Attendance session record
AttendanceReport	student, attendance, status	Per-student attendance entry
StudentResult	student, subject, objective, descriptive, assignment, internal, external, total	Marks entry per exam
Assignment	subject, staff, title, due_date, file	Assignment created by staff
AssignmentSubmission	assignment, student, file, status, marks	Student submission with grade
StudyMaterial	title, subject, file, description	Study materials uploaded by staff
Announcement	title, content, category, audience	Notices for all/staff/students
AcademicCalendar	session, semester, regulation, dates	Semester calendar framework
CalendarEvent	calendar, event_type, start_date, end_date	Specific events in academic calendar
Timetable	course, section, subject, day, period, staff, semester	Weekly class timetable slots

Model	Key Fields	Purpose
Period	number, start_time, end_time	Daily period timing definitions
StudentCloudFile	student, title, file, category	Cloud file storage per student
StudentCertificate	student, title, type, issued_by, issue_date, file	Achievement certificates
Workflow / EmailTemplate	name, trigger_type, graph_data, body	Workflow automation engine

4.2 Database Relationships

- CustomUser ↔ Admin / Staff / Student : OneToOne — each user maps to exactly one role profile
- Student → Course / Session / Regulation / AcademicLevel / AcademicSemester : ForeignKey
- Subject → Course / Regulation / Staff : ForeignKey (subject belongs to a department & regulation)
- Attendance → Subject / Session / Period : ForeignKey (session of attendance)
- AttendanceReport → Student / Attendance : ForeignKey (individual student record)
- StudentResult → Student / Subject : ForeignKey with unique constraint per exam
- Assignment → Subject / Staff : ForeignKey; AssignmentSubmission → Assignment / Student
- Timetable → Course / Subject / Period / Staff : ForeignKey with unique_together constraint
- StudentCloudFile / StudentCertificate → Student : ForeignKey (per-student documents)
- Workflow → WorkflowExecutionLog → Student : audit trail for automation runs

5. Module 1 — Admin / HOD Portal

The HOD/Admin portal is the management control centre of the ERP. It provides comprehensive CRUD operations for all entities and advanced analytics through interactive charts. The admin dashboard is implemented in hod_views.py (103 KB) with 60+ URL routes.

5.1 Dashboard & Analytics

- Live statistics cards: Total Students, Staff, Courses, and Subjects
- Interactive Chart.js visualizations: Subject-wise attendance doughnut chart
- Staff performance analytics with course-wise student count breakdowns
- Recent activity log on the home dashboard
- Quick action shortcuts to all major management sections

5.2 Academic Structure Management

Feature	Description
Degree Management	Create/Edit/Delete B.Tech, M.Tech, and other degree programs
Branch (Course)	Manage departments: CSE, ECE, EEE, ME, CE etc.
Academic Level	Year levels: 1st Year, 2nd Year, 3rd Year, 4th Year
Academic Semester	Semesters within each academic year (e.g., Sem 1, Sem 2)
Session / Academic Year	Annual sessions, e.g., 2023–2024, 2024–2025
Regulation	Syllabus regulations per course (R16, R20, R23 etc.)
Subject Management	Add subjects with code, credits, max marks, staff assignment, ordering
Timetable Builder	Visual weekly timetable with day/period/section/semester
Academic Calendar	Semester-wise calendars with mid-exam, end-exam, and lab exam events

5.3 People Management

- Staff Management — Add, edit, delete staff with full academic and personal profiles (designation, qualification, DOB, Aadhaar, PAN, blood group, experience)
- Staff Qualifications — Multi-entry qualification history per staff member
- Student Management — Comprehensive student profiles with 25+ fields including roll number, section, admission type, blood group, APAAR ID, parent details
- Student Import — Bulk upload via Excel (pandas + openpyxl) with validation
- Student Detail View — 360° student view with marks, attendance, certificates

5.4 Marks & Results

- View consolidated marks report with subject-wise tabulation
- Student Marks Memo — Official formatted marks card per student per semester
- Edit Marks Memo — Admin correction of marks memo before printing
- Add Memo Subject — Add custom subjects to a student's marks memo

- Calculate Final Internal — Auto-compute final internal marks from components
- Export Final Internal Report — Download formatted Excel report per batch

5.5 Communication & Announcements

- Create/Edit/Delete announcements with categories: News, Mid Exams, Semester Exams, Placements, Holidays, Other
- Audience targeting: All users, Staff only, or Students only
- Push notification support via Firebase Cloud Messaging (FCM tokens)
- Feedback review system: Read and reply to student and staff feedback
- Leave approval workflow: Approve/Reject leave applications from staff and students

5.6 Certificate Management

- Add certificates to individual student profiles (Workshop, Sports, Technical, Others)
- View all certificates per student in the student detail page
- Edit or delete certificates with file attachment management

6. Module 2 — Staff / Faculty Portal

The staff portal provides faculty with tools to manage attendance, enter marks, create assignments, upload study materials, and communicate with administration. All operations are scoped to the subjects assigned to the logged-in staff member.

Feature	Details
Staff Dashboard	Personal analytics: subject-wise student count, attendance statistics
Attendance Management	Take attendance per subject/section/period with section filter; update previously entered attendance; grid view by date range
Marks Entry	Add student results: Objective (Quiz), Descriptive, Assignment, Internal, External, and Total marks per subject per exam
Excel Marks Import	Download subject-specific Excel template, fill marks offline, upload for bulk import; download generic template
Export Final Internal	Generate formatted Excel workbook with all marks for reporting
Assignment Management	Create assignments with due date and file attachment; view student submissions; grade and add remarks
Study Materials	Upload PDFs, notes, and documents per subject for student access
Timetable View	View personal class timetable with period-wise schedule
Announcement View	View all announcements published by the HOD
Leave Application	Apply for personal leave with date and reason
Feedback	Send feedback/suggestions to HOD/Admin
Profile Management	View and update personal profile including photo
Notifications	Receive and view admin notifications pushed via FCM

6.1 Attendance System Detail

The attendance module is one of the core features. Staff can take attendance for any subject they are assigned to, selecting the section, semester, date, and period (Period 1–7, Plus period). The system stores attendance per-student per-class and provides a grid view showing daily attendance across a date range with color-coded present (✓) and absent (✗) indicators.

6.2 Marks Entry System

Results are stored in the StudentResult model with separate columns for objective (quiz), descriptive, assignment, internal, and external marks. Staff can enter marks for multiple students at once. The system supports multiple exam entries (Mid-1, Mid-2, etc.) and computes totals automatically. Excel templates can be generated per subject for offline data entry and then imported back.

7. Module 3 — Student Portal

The student portal provides learners with a comprehensive personal academic dashboard. Students can monitor their attendance, access marks and results, view timetables, submit assignments, access study materials, store files in personal cloud storage, and manage their certificates.

Feature	Description
Student Dashboard	Personalized home with attendance summary, quick links, and announcement feed showing latest college notices
Attendance Tracking	View subject-wise attendance summary with percentage; detailed date-wise attendance report with bar/chart visualization
Marks / Results	View marks per exam per subject with component-wise breakdown; consolidated marks table across all semesters
Traditional Results View	Formatted result card with grade and status per subject
Timetable	Personal weekly timetable based on course, section, and semester
Assignments	View active assignments with due dates; submit files for each assignment; track submission status and grades
Study Materials	Access uploaded PDFs and notes from faculty per subject
Cloud Storage	Personal cloud storage with categories: Notes, PDF, Question Paper, Important File; upload/delete files
Certificates	View all uploaded achievement certificates (Workshop, Sports, Technical)
Announcement Board	Browse all active announcements with category filters
Leave Application	Apply for leave with date and reason; track approval status
Feedback	Submit written feedback to HOD/administration
Profile	View personal profile with academic info, contact details
Change Password	Secure password change with current password verification
Notifications	Receive and view admin push notifications

8. Key Advanced Features

8.1 Workflow Automation Engine

A custom **visual workflow builder** (powered by **Drawflow.js**) enables the HOD to create automated email and notification workflows triggered by system events. Workflows are defined as JSON graph data (nodes and connections) and executed by workflow_engine.py.

- Trigger events: New Announcement, Marks Uploaded, Assignment Created, Attendance Alert, Manual Trigger
- Actions: Send email using customizable HTML email templates
- Email templates managed separately with subject and HTML body
- Execution logs stored in WorkflowExecutionLog with success/error status
- Full CRUD for workflows and email templates via admin interface

8.2 Academic Calendar System

Administrators can create detailed academic calendars per session and semester. Each calendar contains structured events with types: Mid-term exams, End-term exams, Lab exams, Branch workshops, Result declarations, and Next semester commencement. Custom events can also be added with free-form naming.

8.3 Excel Import / Export

- Student bulk import from formatted Excel files via pandas
- Subject-specific marks template generation (Excel) for offline entry
- Generic marks template download for new data entry
- Final internal marks workbook export per batch/section
- Excel-based institutional reporting with multi-sheet workbooks

8.4 Cloud Storage for Students

Each student has a personal cloud storage area within the portal. They can upload files of up to any size, categorized as Notes, PDF, Question Paper, or Important File. Files are stored using Django's file storage backend in the student_cloud/ directory under media/.

8.5 Certificate Management

HOD/Admin can add, edit, and delete achievement certificates for individual students. Each certificate record includes: title, type (Workshop, Sports, Technical, Others), issuing authority, issue date, and file attachment. Students can view all their certificates from the student portal.

8.6 Timetable Management

The timetable system supports weekly scheduling per course, section, and semester. Seven periods per day plus additional periods are supported. The system enforces unique constraints (no double-booking of a period for a course/section) and provides both staff and student views of the timetable.

9. Security & Access Control

9.1 Authentication System

- Custom AbstractUser with email as the primary identifier (username removed)
- Email-based login with hashed passwords (Django's PBKDF2-SHA256)
- Session-based authentication with CSRF protection on all forms
- Email validation backend (EmailBackend.py) for authentication
- Password change functionality available for students
- Admin-only creation of user accounts (no public registration)

9.2 Role-Based Access Control (RBAC)

Role	user_type	Access Scope
HOD / Admin	1	Full system access: all CRUD, reports, settings, workflow
Staff / Faculty	2	Own subjects & students: attendance, marks, assignments
Student	3	Own data only: results, attendance, timetable, cloud, certs

9.3 Additional Security Measures

- Right-click context menu disabled across the portal to protect content
- All views protected with @login_required decorator
- Role checks in every view function to prevent horizontal privilege escalation
- CSRF tokens on all POST forms
- Middleware for session management and request authentication
- Media files served through Django (not directly, to allow access control)
- ALLOWED_HOSTS restricted in production settings

10. Deployment & Configuration

10.1 Local Development Setup

Step 1 — Clone	git clone https://github.com/govardhan33699/svrecportalfianl
Step 2 — Virtual Env	python -m venv venv | venv\scripts\activate
Step 3 — Dependencies	pip install -r requirements.txt
Step 4 — Settings	Configure ALLOWED_HOSTS, SECRET_KEY, and database in settings.py
Step 5 — Migrate	python manage.py migrate
Step 6 — Superuser	python manage.py createsuperuser
Step 7 — Run Server	python manage.py runserver → http://127.0.0.1:8000

10.2 Production Deployment (PythonAnywhere)

- WSGI configured via Gunicorn: web: gunicorn college_management_system.wsgi
- WhiteNoise middleware serves all static files in production
- SQLite database file (db.sqlite3) for persistence on PythonAnywhere
- Media files stored in /media/ directory (profile pics, assignments, cloud files)
- Environment variables for SECRET_KEY and database credentials
- runtime.txt specifies Python version for deployment

10.3 Key Configuration Files

File	Purpose
college_management_system/settings.py	Django settings: DB, auth, static, media, installed apps
Procfile	Gunicorn WSGI entry point for production
requirements.txt	All Python package dependencies
runtime.txt	Python runtime version specification
college-erp.yml	Conda environment specification
manage.py	Django management command entry point
db.sqlite3	SQLite database file (1 MB, development)

11. URL Route Map (Summary)

The application has **80+ URL routes** organized by module. Below is a categorized summary.

Common Routes

URL	View	Description
/	login_page	Login page (landing)
/doLogin/	doLogin	Login POST handler
/logout_user/	logout_user	Logout
/get_attendance	get_attendance	Attendance AJAX API

Admin / HOD Routes (Selected)

URL	Feature
/admin/home/	Admin Dashboard
/student/add/	Add Student
/student/manage/	Manage Students
/student/view//	Student Detail View
/student/view//marks-memo/	Student Marks Memo
/student/import/	Bulk Student Import (Excel)
/course_management/	Unified Academic Structure Management
/attendance/view/	Admin View Attendance
/marks/view/	Admin Marks Report
/announcement/manage/	Manage Announcements
/calendar/manage/	Academic Calendar
/timetable/manage/	Timetable Management
/regulation/manage/	Regulation / Syllabus Management
/manage_workflows/	Workflow Automation
/workflow_builder/	Visual Workflow Builder

Staff Routes (Selected)

URL	Feature
/staff/home/	Staff Dashboard
/staff/attendance/take/	Take Attendance

URL	Feature
/staff/attendance/update/	Update Attendance
/staff/result/add/	Add Marks
/staff/result/import-excel/	Import Marks Excel
/staff/result/export-final-internal/	Export Final Internal Excel
/staff/assignment/create/	Create Assignment
/staff/materials/add/	Upload Study Material
/staff/timetable/manage/	View Timetable

Student Routes (Selected)

URL	Feature
/student/home/	Student Dashboard
/student/view/attendance/	View Attendance
/student/view/result/	View Marks/Results
/student/consolidated-marks/	All-semester Consolidated Marks
/student/timetable/	View Timetable
/student/assignments/	View & Submit Assignments
/student/cloud/	Personal Cloud Storage
/student/my-certificates/	View Certificates
/student/announcement/view/	View Announcements
/student/change-password/	Change Password

12. Conclusion & Future Scope

12.1 Conclusion

The **SVREC ERP Portal** is a comprehensive, production-ready college management system that successfully digitizes the entire academic workflow of SVR ENGINEERING COLLEGE. Built entirely with open-source technologies (Python, Django, SQLite), it provides a robust, secure, and user-friendly platform for Administrators, Faculty, and Students.

The system handles 25+ data models, 80+ URL routes, three distinct user portals, advanced features like visual workflow automation, Excel integration, cloud storage, and a certificate management system. The glassmorphism-inspired dark-mode UI ensures a modern, engaging user experience across all devices.

12.2 Key Achievements

- Full role-based ERP with HOD, Staff, and Student portals in a single Django application
- Advanced attendance system with period-wise tracking and grid visualization
- Flexible marks entry with multi-component grading and Excel import/export
- Visual workflow automation engine with email notification support
- Student cloud storage system with category-wise file organization
- Certificate and achievement management per student
- Academic calendar system with structured event types
- Bulk student data import with Excel template generation
- Deployed to production at PythonAnywhere with WhiteNoise static serving
- GitHub repository for version control and collaboration

12.3 Future Scope

Enhancement	Description
Mobile App	React Native / Flutter mobile app consuming Django REST API
REST API	Full Django REST Framework API for third-party integrations
Biometric Integration	Link hardware fingerprint devices to the attendance system
Fee Management	Student fee tracking, payment gateway integration
Library System	Full library management (existing Book model to be expanded)
Parent Portal	Dedicated parent login to view ward's attendance and marks
SMS Notifications	Twilio or local SMS gateway integration for attendance alerts
AI Grade Prediction	Machine learning model to predict student performance
Multi-College	Multi-tenant architecture to serve multiple institutions
PostgreSQL Production	Migrate from SQLite to full PostgreSQL in production
Docker	Containerize the application for easier deployment